

Qingyan (Selena) Lin

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EDUCATION

Graduate Study Plan: Intend to pursue graduate studies in Electrical/Electronics Engineering in Canada after graduation

Georgia Institute of Technology, Atlanta, GA

Expected Graduation: 05/2027

Bachelor of Science in Electrical Engineering, Concentration: Robotics & Signal and Information Processing

Cumulative GPA: 3.69 (Dean's List)

Coursework: Circuit Analysis, Digital System Design, Digital Signal Processing, Control Systems, Microelectronic Circuits, Electromagnetics, Embedded Systems, Autonomous Mobile Robots, Statistics, Computer Vision

Clubs: Electrical Lead, RoboCup RoboJackets – Competitive Robotics | Committee Member, Chinese Student Association

ENGINEERING EXPERIENCES

HSM Solutions, Hickory, NC | *Automation and Machine Vision Intern*

06/2026 - 08/2026

- Developed and validated industrial machine vision inspection systems using LMI vision systems and conveyors, and designed a custom control panel for vision inspection equipment, enabling future optimization.
- Designed and implemented automated inspection pipelines in GoPXL to verify the handrail's hole position and diameter, flat angle offset, bend angle, and surface defects, improving inspection accuracy and consistency.
- Collaborated with engineering teams and vendors to troubleshoot issues, gather requirements, and deliver maintainable, production-ready automation solutions.

RoboJackets @Georgia Institute of Technology, Atlanta, GA | *Electrical Lead – RoboCup Team*

08/2024 - Present

- Designed, built, and validated a custom high-voltage power and discharge circuit, integrating an isolated flyback converter, IGBT switching stage, energy-storage components, and protection circuitry
- Developed and reviewed custom PCBs in KiCad, including schematic capture, component selection, datasheet-driven design, board bring-up, soldering, and hardware debugging
- Led electrical hardware development and integration for the RoboCup 2025 platform, contributing to a 2nd-place finish in the Obstacle Avoidance Technical Challenge, 3rd in Most Improved Team, and 4th overall

Cancer Neurobiology & Nanotechnology Lab, Atlanta, GA | *Research Assistant*

08/2025 - Present

- Enhanced MATLAB scripts to clean and analyze experimental data and generate automated plots
- Developed parametric 3D models in OpenSCAD driven by MATLAB outputs, enabling rapid design iteration and preparation for future 3D printing and prototyping
- Implemented and calibrated resistance measurement procedures and documented experimental methods for repeatability and validation

Georgia Institute of Technology, Atlanta, GA | *Teaching Assistant, Intro to Python*

08/2024 - 05/2025

- Guided students through robotics programming projects, specifically focusing on path planning algorithms, leading to a significant improvement in student performance on related assignments
- Demonstrated strong communication and teaching skills by leading recitations for 50 students, mentoring 5 students one-on-one, and hosting office hours for a class of over 800 students
- Dedicated an average of 10 hours per week to write and grade homework and exam questions with other teaching assistants, demonstrating strong collaboration and cross-functional teamwork skills while maintaining a 3.86/4.0 GPA

ELECTRONICS PROJECTS

Georgia Institute of Technology, Atlanta, GA | *Autonomous Web-Controlled Cart*

04/2026

- Developed a dual-ESP32-C6 robotic cart integrating Wi-Fi control, DC motors, IMU sensing, audio feedback, and OLED indicators.
- Implemented UART communication between microcontrollers and interfaced I²C peripherals including an ICM-20948 IMU and dual SSD1306 OLEDs.
- Controlled DC motors using a TB6612FNG motor driver and PWM, and debugged hardware/software integration issues involving GPIO, serial communication, and motor control.

Georgia Institute of Technology, Atlanta, GA | *LED Brightness and Pattern Control Peripheral*

04/2025

- Designed and implemented a VHDL-based FPGA peripheral for programmable LED brightness and pattern control using PWM.
- Developed FSM logic, clock division, and register decoding for digital control and peripheral operation.
- Performed hardware validation, timing verification, and FPGA debugging to verify functional behavior.

TECHNICAL SKILLS

Programming: Python, C/C++, Matlab, Git/Github, Simulink, ROS, LaTeX, Assembly, VHDL, ssh, Linux

Hardware: Circuit Design, analog electronics, PCB design, Electrical Testing/Debugging, Schematics/layout, ESP32, STM32, FPGAs, Embedded Systems, firmware, UART, SPI, I2C

Instrumentation: Soldering, Multimeter, Oscilloscope, Function Generator, Power Supply, 3D printing, Logic Analyzers

Tools: KiCAD, Intel Quartus Prime, Multisim, CCS, NI LabVIEW, Photoshop, OpenSCAD, LTspice, Inventor, Microsoft Office Suite, Visual Studio Code, AI, Rhino, AutoCAD, Bill of Materials

Certifications: AutoCAD Professional Level II (Autodesk 2021), 3ds Max Professional Level II (Autodesk 2021)